6.6.3 Relation between raw moments and central moments

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu'^3_1$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu_1')^4$$

Example 6.15

The first two moments of the distribution about the value 5 of the variable, are 2 and 20.find the mean and the variance.

Solution:

$\mu'_1 = 2, \mu'_2 = 20$ and $A = 5$

$$\overline{x} = \mu'_1 + A$$

$$\overline{x} = 2+5=7$$

$$\sigma^2 = \mu'_2 - (\mu'_1)^2$$

$$\sigma^2 = 20 - 2^2 = 16$$

Mean = 7 and Variance = 16

6.7 SKEWNESS AND KURTOSIS

There are two other comparable characteristics called skewness and kurtosis that help us to understand a distribution.

6.7.1 Skewness

Skewness means 'lack of symmetry'. We study skewness to have an idea about the shape of the curve drawn from the given data. When the data set is not a symmetrical distribution, it is called a skewed distribution and such a distribution could either be positively skewed or negatively skewed.



The concept of skewness will be clear from the following three diagrams showing a symmetrical distribution, a positively skewed distribution and negatively skewed distribution.

We can see the symmetricity from the following diagram.

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